

Landscape-scale bacterial biogeography across the Rub' al-Khali

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ABSTRACT

The Rub' al-Khali (Empty Quarter) is the world's largest continuous sand desert, yet the bacterial ecology of its open Saudi interior has not been surveyed. We present the first landscape-scale bacterial survey of this region, with 5 expeditions to 60 sites over 31 months and paired surface, shallow-subsurface, and root-adjacent soil. Bacterial communities formed a recurring geographic pattern: distant sites contained more different communities, mainly because taxa replaced one another across the landscape. Within sites, root-adjacent communities were distinct and less even than both bulk soil compartments, showing that local soil context changes which bacteria dominate. Long-term temperature, rainfall, and humidity tracked diversity and the relative abundance of many genera, while richness showed a short positive association with recent rainfall, strongest during the first several days. Soil pH accounted for part of the geographic pattern. The metabolic pathways predicted from the 16S sequences retained both geographic and compartment structure, and their broad gene-family profiles agreed with matched shotgun data. The main ecological conclusions held under checks for DNA from damaged cells and for low-biomass contamination. The study establishes a regional biological baseline and a reusable resource for biodiversity monitoring, restoration, arid-agriculture research and evidence-based land planning.

Keywords: Rub' al-Khali, Empty Quarter, hyperarid soil microbiome, bacterial biogeography, landscape-scale survey, spatiotemporal, repeated sampling, 16S rRNA amplicon survey

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INTRODUCTION

The Rub' al-Khali (Arabic الربع الخالي; in English, the Empty Quarter) is the world's largest continuous sand desert. It covers much of the southern Arabian Peninsula, mainly within Saudi Arabia [1], and shapes land use, conservation, and plans for agriculture across the region. Yet the bacteria in its open interior have not been surveyed. The closest molecular community study examined a groundwater-fed salt flat in the United Arab Emirates, a habitat unlike the open sand soils of the Saudi interior [2]. A regional survey is needed both to understand this major desert and to provide a reference for future monitoring, restoration and arid-land research.

Work in other deserts shows why such a survey matters. Dry landscapes that look uniform can contain strong changes in bacterial composition. Climate, soil pH, salts and minerals are often associated with these changes [3, 4, 5, 6]. Rare rainfall can also trigger a short biological pulse before the soil dries again [7, 8]. Sampling sites at different delays after rainfall can therefore test whether diversity rises and then declines. One visible spatial outcome is distance decay: nearby sites contain more similar communities than distant sites [9, 10, 11]. This pattern can arise because taxa replace one another across the landscape, or because poorer communities become subsets of richer ones. Replacement implies distinct local assemblages; nestedness implies loss from a shared pool. Neither process has been measured across the open Empty Quarter.

Desert soil can also change within centimetres. The surface receives direct sunlight and strong heating, while soil at 5–10 cm is partly buffered [12]. Sparse plants may add water and carbon near their roots [13, 14]. These differences can change which taxa occur, which taxa dominate, and which metabolic pathways the community is predicted to encode. Separating these outcomes requires abundance and functional measures alongside richness: two soils may contain similar numbers of taxa while their abundance and predicted functional profiles differ. DNA can also remain after cells die. This relic DNA, together with contamination introduced during laboratory processing, is a particular concern in low-biomass desert soil. Biological interpretation therefore requires controls that distinguish these sources of DNA from the sampled community [15].

Surveys elsewhere on the Arabian Peninsula have described bacteria along elevation gradients, at single arid locations, and in plant-associated soil [16, 17, 18, 13, 19, 14, 20]. None covers the open Empty Quarter at landscape scale or combines a dense transect with repeated sampling of several compartments. We therefore sampled 60 sites during 5 expeditions over 31 months. We asked how bacterial composition changes across the landscape and among compartments; how climate, pH and elemental composition relate to those patterns; whether predicted metabolic potential changes with them; and whether relic-DNA and low-biomass controls alter the main conclusions. The resulting survey maps bacterial biogeography at landscape and centimetre scales, with the samples, measurements, sequences, and their production records preserved as a linked resource for reuse [21]. Communities form a recurring geographic mosaic in which taxa replace one another across the sand sea. Within a site, the compartment (surface, shallow-subsurface or root-adjacent soil) changes which taxa dominate rather than how many taxa are detected. Long-term climate and soil pH track the geographic pattern. Richness shows a short positive association with recent rainfall, and predicted metabolic potential retains both spatial structures; neither the relic-DNA nor the low-biomass controls change these conclusions.

RESULTS

Bacterial communities change across the landscape

We surveyed 60 sites along a west–east transect through the Saudi Empty Quarter and returned during 5 expeditions over 31 months (Figure 1a). At each visit, we sampled bulk surface soil, bulk subsurface soil (“deep” soil), and root-adjacent soil, collecting three replicates per compartment. We sequenced the V3–V4 marker region of the bacterial 16S rRNA gene and resolved the reads into amplicon sequence variants (ASVs). This yielded 1,237 quality-controlled profiles, of which 1,227 come from the repeatedly sampled sites 1–60 (Figure 1b). Trip 2 covered only 8 sites due to extreme heat. Supplementary Table S1 lists the samples used in each analysis.

Eight phyla each held more than 2% of reads on average across the 1,227 core-site profiles: Pseudomonadota (26.9%), Bacillota (22.5%), Actinomycetota (18.7%), Chloroflexota (8.6%), Bacteroidota (7.1%), Planctomycetota (3.2%), Gemmatimonadota (2.7%) and Acidobacteriota (2.3%); Cyanobacteriota held 0.4% (Supplementary Section S5.6). Bacilli was the leading class (21.0%), ahead of

Gammaproteobacteria (14.7 %), Actinobacteria (13.3 %) and Alphaproteobacteria (12.2 %). The leading genera were *Domibacillus* (6.2 % of reads), *Massilia* (3.5 %), *Bacillus* (3.0 %), *Microvirga* (2.4 %) and *Flavisolibacter* (2.0 %), followed by *Noviherbaspirillum*, *Streptomyces*, *Herpetosiphon*, *Paenibacillus* and *Lysinibacillus*; *Domibacillus* led every compartment. Endospore-forming Bacillota and Actinomycetota therefore supplied over 40 % of reads, whereas Acidobacteriota and Verrucomicrobiota, which rank among the most abundant phyla in global soil surveys [4, 22], together held 3.3 %. Reads without a genus assignment made up 35.6 % of the total, archaea 0.02 %, and the Cyanobacteriota were no more abundant at the surface (0.40 %) than below it (0.41 %) or near roots (0.51 %), so the bulk soil sampled carried no biocrust signature.

We investigated whether community composition changed gradually across the sampled transect. The model allowed two broad patterns: a steady change from west to east, or a smooth change in its rate that could reverse direction once. These were represented by a straight-line term and a squared-location term. Together, these patterns accounted for 40.1 % of the site-averaged difference in relative taxon abundance across 630 site, expedition, and compartment combinations (95 % CI, 32.4–47.7 %; $p = 0.001$). The result remained when we changed the number of genera, used ASVs as the taxonomic units, or omitted any one expedition (Supplementary Section S5.3). The geographic pattern therefore held at both taxonomic resolutions and with every expedition omitted in turn.

Communities also became less similar with distance. We therefore asked a more direct question: did sites farther apart have more different relative taxon abundances? We measured this with Aitchison dissimilarity, which compares the ratios among taxa. It is zero when two relative compositions are identical and increases as their taxon ratios diverge [23]. A positive slope with distance therefore shows how rapidly community composition changes across the landscape. The slope was 1.349 (95 % CI, 0.903–1.795), 1.507 (1.089–1.925) and 1.175 (0.695–1.656) dissimilarity units per 100km in surface, shallow-subsurface and root-adjacent soil, respectively (each $p = 0.0001$; Figure 1c). The slopes differed among compartments ($p = 0.0051$): composition changed fastest below the surface and slowest next to roots. Genus occupancy followed the same order. After pooling expeditions within site and subsampling every site to 12,865 reads, root-adjacent soil had 85 genera present at 90 % or more of the sites, surface soil 82 and shallow-subsurface soil 69, and the median number of genera per site was 299, 274 and 284 (Supplementary Section S5.7). The compartment with the largest shared pool changed most slowly with distance.

Finally, we asked what kind of change produced these differences. After comparing every sample at the same read coverage, we used Sørensen dissimilarity to compare the proportions of shared and unshared taxa between two sites. It is zero for identical taxon lists and increases as fewer taxa are shared. We separated taxa that replaced one another between sites from taxa lost from an otherwise shared pool; these components are called turnover or replacement and nestedness, respectively [24]. Replacement represented 69–75 % of the average difference in every compartment. Both components rose with distance, but replacement dominated. The Empty Quarter is therefore a mosaic of distinct bacterial assemblages that replace one another along the route.

The replacement had identifiable participants. Among the 200 genera of the route model, 124 changed monotonically along the route after correction for the 200 tests: 77 declined from west to east and 47 increased (Supplementary Section S5.6). The declining genera were mostly Actinomycetota (27 of 77; for example *Cellulomonas*, Spearman $\rho = -0.79$, *Marmoricola* and *Streptomyces*), Pseudomonadota (20; *Sphingomonas*, *Steroidobacter*, *Massilia*), Bacteroidota (9) and Acidobacteriota (4; *Blastocatella*, *Aridibacter*). The increasing genera were led by Bacillota (17 of 47): *Bacillus* rose from 1.6 to 6.2 % of reads between the western and eastern thirds ($\rho = 0.77$) and became the leading genus of the eastern third, joined by genera typical of saline and alkaline soils such as *Halalkalibacter* ($\rho = 0.85$), *Halomonas*, *Aquibacillus* and *Pseudalkalibacillus*. The eastern sites are also the warmer, more humid and more alkaline end of the route (site means: Spearman ρ with route position 0.98 for temperature, 0.92 for humidity and 0.82 for pH), so these genera mark the environmental gradient.

The pattern is a property of the dune soils. Field notes assign each site one landform: 44 sand-dune sites, 9 saline pans (3 in the west, 1 central, 5 in the east), 3 oases in the east, and one gravel plain, dune slack, aeolian lake and oil-spill site each (Supplementary Section S5.7). Saline-pan sites differed from dune sites in composition before adjustment for route position (pseudo- $F = 4.56$, $p = 0.0013$; 16 genera at $q < 0.05$) but barely after it (pseudo- $F = 1.74$, $p = 0.036$; no genus at $q < 0.05$), and their diversity matched that of dune sites ($p \geq 0.28$). The three oasis sites had the lowest diversity (mean Shannon 4.25

against 5.77 at dune sites). On the 44 dune sites alone, the route model explained 34.2% of composition (40.1% on all sites), 70 of the 71 genera with a supported route correlation on dune sites changed in the same direction as on all sites, and the temperature and humidity associations with diversity remained supported. The west-to-east replacement is therefore a property of the dune soils themselves.

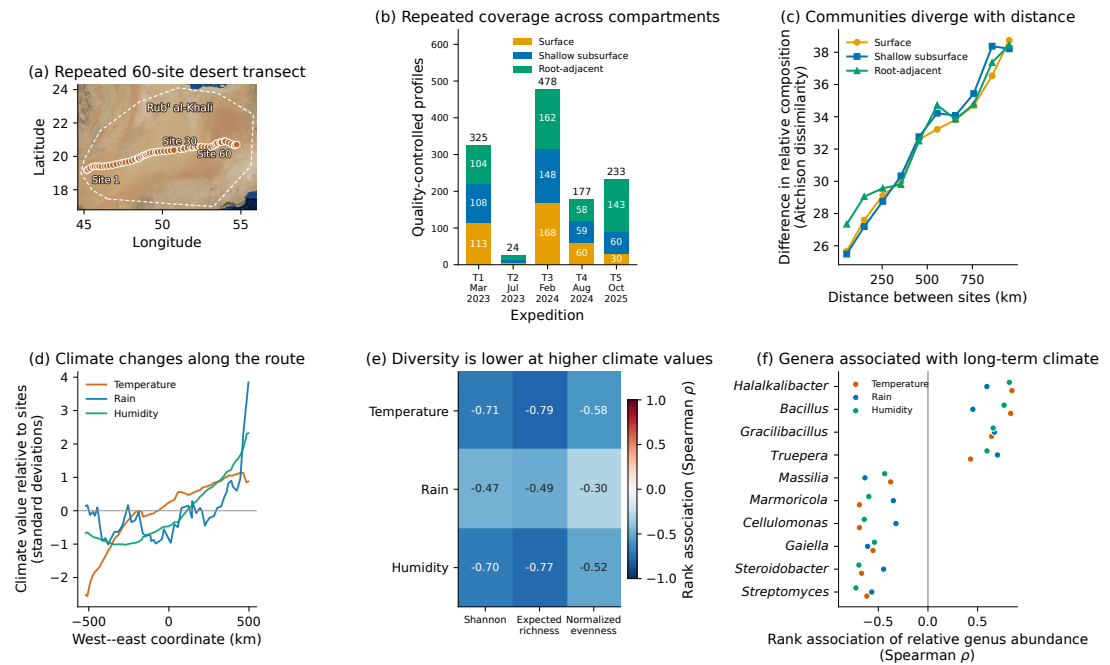


Figure 1. Survey design, geographic differences and climate associations. (a) The 60 core sites along the west–east route on a NASA Blue Marble satellite image (July 2004); the dashed outline marks the Empty Quarter boundary for orientation only. (b) The 1,237 quality-controlled profiles by expedition and compartment. Trip 2 reached 8 sites. (c) The difference in relative community composition, measured as mean Aitchison dissimilarity, as distance between sites increases. The statistical test kept all comparisons involving the same site together. (d) Each site’s 49-month climate mean, expressed as standard deviations from the mean across all sites. (e) Rank associations between climate and diversity. Values near -1 mean that diversity consistently decreases as the climate variable increases; values near zero mean no consistent order. (f) The strongest supported genus associations. Tests in (e,f) were corrected for the number of comparisons.

Compartment shapes bacterial composition and evenness

We next asked whether compartment changed the relative abundance of taxa within the same site and expedition. Across 170 site-by-expedition comparisons in which all three compartments were available, the separation among the three average compositions was large relative to the variation within compartments. This ratio is the pseudo- F statistic: a larger value means stronger overall separation. The compartment separation had pseudo- $F = 17.28$ ($p = 0.001$). We then asked whether sites shifted in a consistent direction for each compartment pair. We divided the magnitude of the average shift by the average magnitude of the individual site shifts. This standardized displacement is near 1 when sites shift in the same direction and near 0 when their shifts cancel. Root-adjacent and shallow-subsurface soil showed the most consistent shift (0.478), followed by root-adjacent and surface soil (0.433) and shallow-subsurface and surface soil (0.331; Figure 2a). All 3 comparisons passed correction for the 3 tests and remained supported after changing the included taxa, the treatment of zero counts, or the omitted expedition (Supplementary Section S5.1).

Shannon diversity combines the number of taxa with the balance of their read abundances. It was lower in root-adjacent than shallow-subsurface soil by 0.206 on average (Figure 2b). After resampling whole sites, the 95% interval was $[-0.398, -0.001]$. The difference remained after adjustment for sequencing depth and recorded extraction method (-0.273 , 95% CI $[-0.424, -0.122]$).

To find what produced this change, we estimated how many taxa would be expected if every profile had 25,000 reads. We then scaled Shannon diversity by this common-depth richness to separate the balance among taxa from their number. This measure is normalized evenness: a high value means that reads are spread across taxa, whereas a low value means that a few taxa dominate. Evenness declined from shallow-subsurface to surface and then to root-adjacent soil (Figure 2c). Root-adjacent soil was lower than surface soil by 0.0315 (95 % CI [−0.0448, −0.0186]) and lower than shallow-subsurface soil by 0.0440 ([−0.0572, −0.0313]). Expected richness ordered the compartments differently. Compartment therefore changed which taxa dominated more consistently than it changed how many taxa were detected. Because the read abundances are relative, we compared each genus with the geometric mean abundance across all genera on a logarithmic scale. This avoids treating the reads as absolute cell counts. Testing the 200 route-model genera in each of the 3 paired contrasts (600 tests) supported 295 differences after correction (Supplementary Section S5.1). Surface soil held more *Deinococcus*, *Kineococcus*, *Kocuria*, *Planococcus* and *Rufibacter* than shallow-subsurface soil and more *Deinococcus*, *Rubrobacter*, *Geodermatophilus* and *Blastococcus* than root-adjacent soil, genera known for pigmentation and tolerance of radiation and desiccation. Root-adjacent soil held more *Pelagibacterium*, *Neorhizobium*, *Devosia*, *Metabacillus*, *Pseudomonas* and the Saccharibacteria lineage TM7a than either bulk compartment, and more *Brevundimonas* and *Pantoea* than shallow-subsurface soil; *Domibacillus*, the leading genus overall, was also highest near roots. Shallow-subsurface soil held more of the nitrifiers *Nitrospira* and MND1 (Nitrosomonadaceae) than either other compartment, and more *Gaiella*, *Symbiobacterium* and *Caldilinea* than root-adjacent soil. *Bradyrhizobium*, a root-nodule genus, was lower near roots than in either bulk compartment (Figure 2d). These contrasts describe which genera carry the compartment difference.

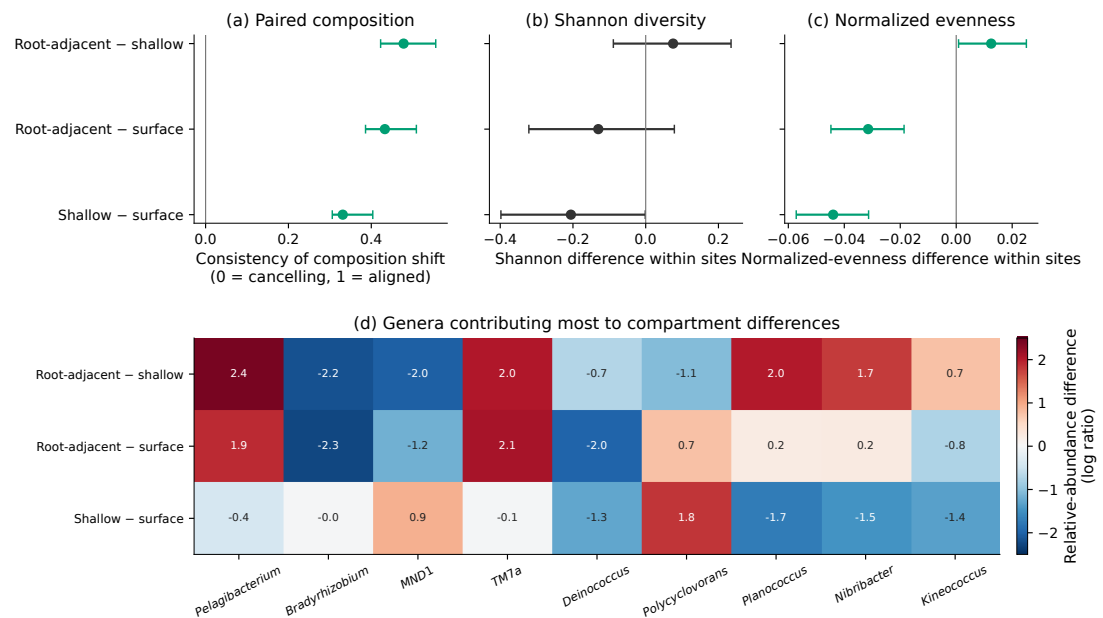


Figure 2. Compartment differentiates bacterial communities within sites. (a) The consistency of the shift in relative taxon abundance across sites. Values approach one when sites shift in the same direction and zero when shifts cancel. (b,c) Paired differences in Shannon diversity and normalized evenness, averaged across expeditions. In (a–c), points show means and bars show 95 % intervals. (d) Genera that contributed most to the 3 overall composition differences. Positive values mean a higher relative abundance in the first named compartment. This panel describes contributions to the overall difference.

Climate and soil properties track bacterial variation

We recorded air temperature, pressure and relative humidity during 247, 177 and 239 quality-controlled site visits. Air temperature ranged from 20.0 to 49.4°C and humidity from 9.5 to 62.5 %. To describe background climate, we also summarized 49 months per site from January 2022 to January 2026. Long-term site means ranged from 26.38 to 30.30°C, 1.31 to 7.47 mm rainfall per month, and 24.27 to 35.40 %

relative humidity. Figure 1d shows how these site means changed across the transect.

We compared 3 aspects of diversity: Shannon diversity combines taxon number and abundance balance, expected richness estimates how many taxa would be detected at a common read coverage, and normalized evenness measures how evenly reads are spread after accounting for richness. We then asked whether sites ordered from lower to higher climate values also showed a consistent order in each. Spearman rank correlation measures this ordering: a value near -1 means that diversity consistently decreases as the climate value increases, whereas a value near zero means no consistent order. All 3 long-term gradients tracked all 3 measures. Sites with higher temperature, rainfall and humidity had lower Shannon diversity (Spearman $\rho = -0.71, -0.47$ and -0.70), lower expected richness ($-0.79, -0.49$ and -0.77), and lower evenness ($-0.58, -0.30$ and -0.52) (Figure 1e). All 9 associations remained supported after correction for the 9 comparisons (adjusted $p \leq 0.021$). Among 200 genera detected in at least 20% of profiles, 112, 112 and 111 were associated with temperature, rainfall and humidity after multiple-testing correction (Figure 1f). The strongest positive associations with temperature were *Halalkalibacter*, *Bacillus*, *Pannonibacter*, *Ammoniphilus* and *Rhodocista* ($\rho = 0.74-0.85$), and the strongest negative associations were *Cellulomonas*, *Marmoricola*, *Steroidobacter*, *Ellin6055* and *Pseudonocardia* ($\rho = -0.65$ to -0.69); *Streptomyces* had the strongest negative association with humidity ($\rho = -0.73$). Because climate covaries with location along the route, these are the genera that mark the west-east replacement described above. Supplementary Section S7 gives the genus rankings and shows how strongly these climate patterns overlap with location.

Quality-controlled pH measurements matched 702 profiles and had a mean pH of 7.986. After accounting for expedition and compartment, location along the route explained 84.6% of pH variation ($p \leq 0.001$). Among the same 702 matched profiles, the share of bacterial composition explained by location fell from 35.8% to 19.6% after pH adjustment. Excluding the group with the highest pH, or its entire site, gave 18.7–19.6%. pH and location therefore described substantial overlapping variation. The genera that tracked site pH were the genera of the route: 117 of the 200 genera correlated with site-mean pH ($q < 0.05$), 104 of them also with route position. *Polygonibacillus*, *Sediminibacillus*, *Halalkalibacter*, *Gracilibacillus*, *Halomonas* and *Aquibacillus* increased with pH ($\rho = 0.71-0.81$); *Pirellula*, *Roseisolibacter*, *Gemmatimonas*, *Steroidobacter*, *Gaiella*, *Sphingomonas* and *Streptomyces* decreased ($\rho = -0.60$ to -0.68 ; Supplementary Section S5.7).

X-ray fluorescence (XRF) measured 11 elements in at least 75% of 725 specimens. Because the elements changed together, we combined them into one summary gradient that captured the largest shared change. This gradient represented 39.2% of the elemental variation and a smaller but consistent share of bacterial composition, 0.358% ($p = 0.010$). Planned changes to the included taxa and treatment of zero values gave estimates from 0.311 to 0.543% (Supplementary Sections S2–S3). This range shows sensitivity to analysis choices. The gradient loaded positively on Ca, Mg, Na, S, Cl, Fe and Ti and negatively on Si, so it separates sand rich in carbonate, sulfate and chloride minerals from quartz-rich sand; it followed the route (Spearman $\rho = 0.73$ with route position). 118 genera tracked it, with the same *Bacillota* (*Polygonibacillus*, *Sediminibacillus*, *Aquibacillus*, *Halalkalibacter*, *Gracilibacillus*) on the Ca, Mg, Na, S and Cl side and *Steroidobacter*, *Caldilinea*, *Streptomyces*, *Roseisolibacter* and *Gemmatimonas* on the Si side (Supplementary Section S5.7).

Bacterial richness shows a short association with recent rainfall

We next asked whether rare rainfall was followed by a temporary rise in diversity. The expeditions ran on fixed dates, so each sample was collected an incidental number of days after the last rainfall at its site, and the delay at which any response peaks was unknown. For each sample we therefore weighted rainfall over the preceding 60 complete days by a curve that rises to a maximum at day p and then decays, so that rainfall p days before collection carries the most weight. We scanned p from 0.5 to 30 days across 3 diversity measures (180 candidate models) while accounting for site, compartment, expedition and location along the route, and retained the strongest positive association.

Retaining the best of 180 results inflates significance, so we calibrated the complete search. We rotated each expedition's 60-day rainfall series in a circle 19,999 times, independently within expeditions. Each rotation preserves the amount, rarity and clustering of rainfall and its spatial pattern across sites, and breaks only its timing relative to collection; because rotation acts within an expedition, the calibration holds when storms occur in only some expeditions. Re-running the entire scan on every rotation gave the distribution of the largest statistic obtainable when rainfall timing is uninformative, against which we

tested the observed result.

Expected richness was the selected endpoint in both daily rainfall products. With NASA POWER, the primary series [25], each additional millimetre at the fitted peak was associated with 179.6 additional expected taxa (95 % site-clustered CI, 92.1–267.1); the fitted peak was 2 complete days (95 % peak-selection interval, 1.5–4.5 days), rainfall accounted for 3.64 % of the remaining variation (0.60–7.87 %), and the family-wise one-sided test for a positive pulse gave $p = 0.0560$. Open-Meteo, an independent product [26], selected the same endpoint with a peak at 1 day (0.5–6.5 days), a larger estimate of 327.5 taxa per millimetre (141.1–514.0), 3.84 % explained variation (0.57–8.99 %), and $p = 0.0432$ after the same correction. The two products therefore agree on an early positive association but differ in its size and exact timing (Supplementary Figure S1).

Three checks bound this result. The pulse curve assumes a smooth rise and decline; fitting rainfall from 7 non-overlapping periods before collection instead, with no assumed shape, placed the largest positive estimate at 3–4 days in both products. Rain falling after collection, which cannot have an effect, produced no comparable pattern (largest richness statistic 0.055). The association was positive in every compartment and changed little after accounting for pH, but weakened when the model allowed more complex geographic change. Removing candidate contaminants also left the estimate essentially unchanged, although its corrected probability then crossed the conventional threshold ($p = 0.0485$); we therefore report the unfiltered estimate. The data support an early association between rainfall and richness (Supplementary Section S7).

Predicted metabolic pathways follow geography and compartment

We used PICRUSt2, which compares each 16S sequence with reference genomes, to predict which metabolic pathways the community could encode [27]. These estimates describe metabolic potential. Across 1,227 core-site profiles and 462 pathways in the MetaCyc database [28], the predicted profiles were dominated by housekeeping metabolism. Biosynthesis pathways held 66.8 % of predicted pathway abundance, energy and central metabolism 17.2 % and degradation or utilization pathways 10.7 % (key-word classes; Supplementary Section S8.1). Aerobic respiration through cytochrome *c* was the most abundant single pathway (1.7 %), followed by branched-chain amino acid and fatty acid biosynthesis. Carbon fixation was minor: the Calvin–Benson–Bassham cycle ranked 44th (0.6 %) and the reductive TCA cycle 122nd (0.4 %), in line with the 0.4 % read share of Cyanobacteriota; the MetaCyc set used by PICRUSt2 contains no light-reaction pathway. The predicted potential therefore describes an aerobic, heterotrophic community. Against this shared background, the 200 most abundant pathways retained a strong geographic pattern. The same route model, which allowed the rate of change to vary smoothly and reverse once, accounted for 23.3 % of the difference among site-level pathway profiles (95 % CI, 15.4–31.3 %; $p = 0.0001$). The result remained supported after changing the included pathways, the treatment of zero values, or the omitted expedition.

We then asked whether predicted pathway profiles differed among compartments, and found a difference across 172 complete site-by-expedition comparisons ($p = 0.0001$). We used the same standardized displacement as for taxonomic composition. Predicted pathways shifted most consistently between root-adjacent and shallow-subsurface soil (0.453) and between root-adjacent and surface soil (0.429; both adjusted $p = 0.00015$ after correction for 3 comparisons; Figure 3a). Of 600 tests covering 200 pathways and 3 compartment pairs, 270 remained supported after correction. Root-adjacent soil had the largest increases for formaldehyde assimilation through the ribulose monophosphate cycle, palmitate and lysine biosynthesis, menaquinol biosynthesis, and glucose, xylose and fructuronate degradation; shallow-subsurface soil for isoleucine biosynthesis IV, the incomplete reductive TCA cycle, GDP-mannose biosynthesis and starch degradation; and surface soil for alanine and biotin biosynthesis and tRNA processing. Supplementary Section S8.1 gives the complete results.

From west to east along the transect, 2 pathways that produce the fatty acids *cis*-vaccenate and gondoate increased, while palmitate production decreased. Root-adjacent soil had higher predictions for glucose and xylose degradation and for several polyamine and thiamin pathways. These patterns identify candidates for tests of membrane adaptation and carbon and metabolite exchange near desert roots.

We then asked a narrower question of a fixed list of 20 trait genes that matter in hyperarid soil, using two independent gene sources (Supplementary Section S8.3). In the 975 annotated genomes of the companion shotgun catalogue, 38.5 % carried the aerobic CO dehydrogenase *coxL* and 22.6 % a group 1 [NiFe]-hydrogenase large subunit, most of them Actinomycetota; 24.6 % carried RuBisCO,

0.2 % photosystem II and 1.0 % nitrogenase (*nifH*). Osmolyte synthesis was widespread (trehalose in 70–73 % of genomes, ectoine in 36 %), as were carotenoid synthesis (49 %), DNA repair (*uvrA* 95 %, *recA* 92 %) and oxidative-stress genes (catalase 77 %, superoxide dismutase 83 %). Weighted by genome abundance in the 150 shotgun libraries, hydrogenase carriers made up 32 % of the recruited community in shallow-subsurface soil against 17 % at the surface and 15 % near roots, whereas carotenoid and photolyase carriers were most abundant at the surface (61 and 34 %). The PICRUSt2 predictions agreed with the genome-derived shares for the trace-gas, ectoine, trehalose (*otsA/otsB*), sporulation, catalase, superoxide dismutase and Dps traits (Spearman $\rho = 0.51$ – 0.76 across 119 matched libraries) and gave the same compartment pattern: predicted hydrogenase and *coxL* shares were lowest near roots (0.67 and 0.74 of the shallow-subsurface value), whereas sporulation (*spo0A*, 1.4–1.5-fold), catalase, superoxide dismutase and Dps were highest there. Along the route, predicted trehalose-synthesis and carotenoid genes decreased eastward ($\rho = -0.71$ and -0.59) while ectoine, *spo0A* and superoxide dismutase genes increased (0.36, 0.56 and 0.70). Gene presence describes potential, and the hydrogenase KOs pool several [NiFe]-hydrogenase groups.

Prediction quality depends on how closely the sampled bacteria resemble organisms with sequenced reference genomes. PICRUSt2 summarizes this distance as the weighted nearest-sequenced-taxon index (NSTI); smaller values require less extrapolation, and the median here was 0.165. We also made a more direct comparison in 125 samples. For each sample, we ranked 3,471 gene families by abundance in the PICRUSt2 prediction and in shotgun metagenomic data. Shotgun sequencing reads all recovered DNA, so it measures gene families directly. These families, called KEGG Ortholog groups, collect genes with the same broad function. A rank correlation near one means that the two methods place the gene families in nearly the same order. The median correlation was 0.743 (95 % interval, 0.731–0.751, after resampling whole sites), and the correlation between the two community-average profiles was 0.806 (0.793–0.812; Figure 3b). This level of agreement supports broad functional comparisons while leaving individual pathways as predictions [27, 29]; Supplementary Section S8.2 gives the matched-data analysis.

DISCUSSION

The Empty Quarter and global dryland ecology

The open soils of the Saudi Empty Quarter are organized both across hundreds of kilometres and within a few centimetres. Taxa replace each other throughout the landscape, while compartment changes the composition of the community and the balance between taxa within the sites. Climate, pH, and elemental composition place this variation in an environmental setting, and predicted pathways suggest that taxonomic changes carry differences in metabolic potential. The main lesson is that a hyperarid sand sea contains a repeatable mosaic at landscape and local scales.

The phylum profile places the Empty Quarter with other hot deserts. Pseudomonadota, Bacillota, Actinomycetota and Chloroflexota led, and Acidobacteriota and Verrucomicrobiota, prominent in temperate and agricultural soils, were minor [30, 22, 4]. The genus level adds a desert-specific signature: endospore-forming *Domibacillus*, *Bacillus*, *Paenibacillus* and *Lysinibacillus*, together with *Massilia*, *Microvirga* and *Streptomyces*, supplied more reads than any other genera.

The change in the geography of the community recurs in desert microbiomes. In the Empty Quarter, the transect pattern remained at genus and ASV resolution and across expeditions. Taxon replacement formed 69–75 % of mean dissimilarity, so distant sites usually held different assemblages. The replacement has a recognizable biology: Actinomycetota, Acidobacteriota and Gemmatimonadota genera such as *Cellulomonas*, *Marmoricola*, *Blastocatella* and *Roseisolibacter* gave way eastward to Bacillota, with *Bacillus* and genera typical of saline and alkaline soils gaining most, in the direction in which temperature, humidity and pH increased. The replacement is a property of the dune soils: the same genera changed in the same direction on the 44 dune sites alone, and saline pans differed little from dunes once route position was accounted for. Distance decay also occurs in Negev soils [11], and Namib dune systems show habitat-specific community structure across dune zones [31]. Different primers, distances and soils prevent a merged numerical comparison, but a presence-based comparison bounds the question of whether the Empty Quarter is one ecosystem or several. After subsampling every profile to 8,000 reads and requiring detection in at least 10 % of profiles, the western and eastern thirds of the transect shared 267 of 418 genera (Jaccard 0.64; 0.68 and 0.83 for adjacent thirds), and all 50 leading genera of either end were detected at the other. The Empty Quarter and a public Atacama soil pit [12] shared 43 of 397 genera (Jaccard 0.11), and only 17 of the 50 leading Empty Quarter genera reached the detection

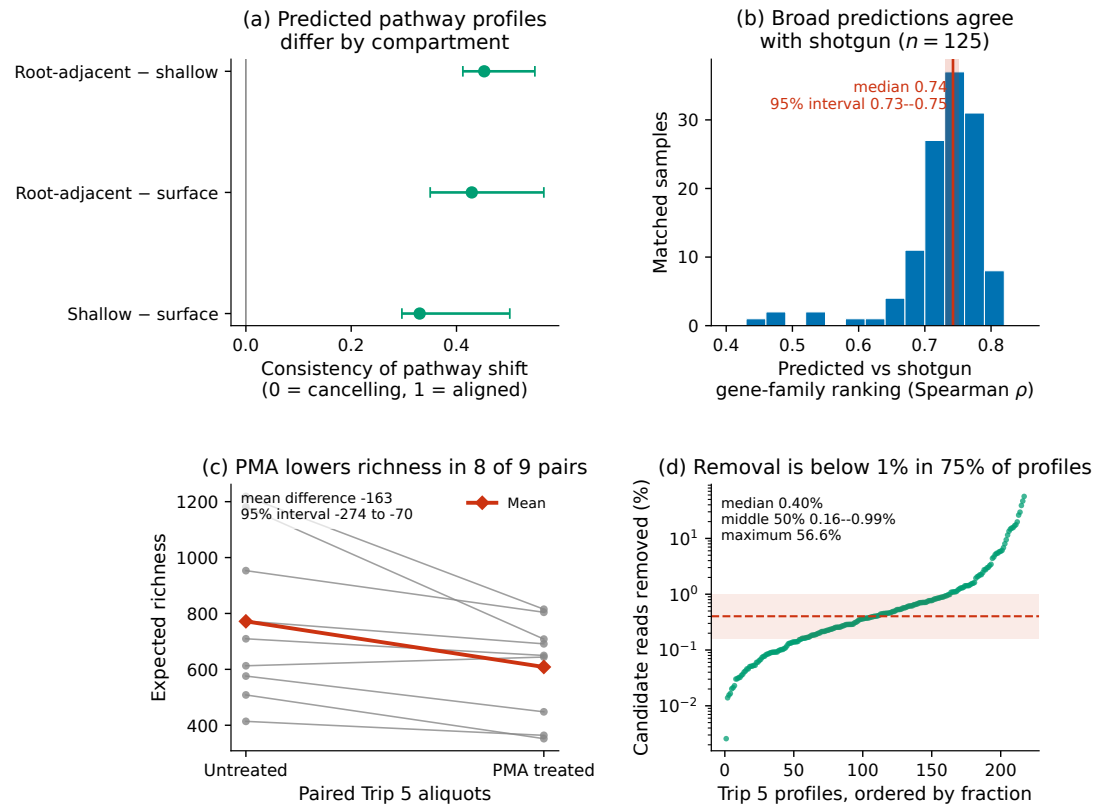


Figure 3. Predicted function and assay controls. (a) Paired differences in predicted pathway profiles. Values approach one when sites shift in the same direction and zero when shifts cancel; bars show 95% intervals from resampling whole sites. (b) Agreement between predicted and shotgun-measured gene-family rankings; red marks the median and its 95% interval. (c) Estimated richness before and after PMA treatment; red joins the means and the annotation gives the interval for their difference. (d) The fraction of reads assigned to candidate contaminants; red marks the median and the range containing the middle half of profiles.

threshold in the Atacama profiles (Supplementary Section S9.2). The overlap between the transect ends was therefore six times the overlap between deserts. Within-desert replacement is strong, but it reorders a largely shared regional pool; the Empty Quarter behaves as one regional system with internal structure. Primer, DNA-fraction and depth differences inflate the Atacama contrast, and all three deserts reject the hypothesis that there is one representative “desert soil” community.

Water was associated with diversity on two scales. Along the landscape, long-term temperature, rainfall, and humidity were negatively associated with diversity. After individual rainfall events, by contrast, richness was highest during the first several days, and the fitted association then declined. These patterns answer different questions: the long-term comparison separates sites with different climate histories, while the pulse model compares rainfall amount and time before collection. Pulse-reserve theory predicts that rare rainfall can briefly release water limitation before drying restores it [7, 8]. Rapid cell resuscitation, solute movement, or improved detection of rare taxa could each produce our pattern; process measurements after rainfall would separate them. Direction also varies among deserts. In a public Atacama gradient, soil humidity was positively associated with Shannon diversity even after accounting for latitude, longitude, and elevation (rank correlation 0.68, $p = 0.011$) [32]. Therefore, water source, evaporation, soil chemistry, and the range sampled matter more than a global rule such as “more water means more diversity” [33].

Physical stratification also occurs, although its direction depends on scale. An Atacama pit showed differences in community composition among depth zones ($p = 0.0014$) without a steady increase or decrease in diversity, and Mojave biocrust communities differ between surface and subsurface soil [12, 34]. The Empty Quarter adds repeated comparisons of surface, 5–10 cm and root-adjacent soil. Root-adjacent communities were less even at similar richness, so fewer taxa carried more reads. The genera behind this pattern form three recognizable guilds. The surface is enriched in pigmented, radiation- and desiccation-tolerant Actinomycetota and Deinococcota (*Deinococcus*, *Rubrobacter*, *Geodermatophilus*, *Blastococcus*, *Kocuria*), the shallow subsurface in nitrifiers (*Nitrospira*, Nitrosomonadaceae MND1; the genome catalogue also holds ammonia-oxidizing Thermoproteota) and in slow-growing *Gaiella*, and root-adjacent soil in copiotrophic Rhizobiales (*Pelagibacterium*, *Neorhizobium*, *Devosia*), *Pseudomonas* and Bacillota, with the highest predicted sporulation and oxidative-stress gene shares and the lowest trace-gas oxidation potential. This pattern is consistent with a stronger selective environment near roots: plants change carbon and nutrient supply, and studies of desert plants have found host- and root-compartment-specific recruitment [35, 13, 36, 20]. Proximity to a root defines this compartment, so the root is the most likely source of the difference. The host plants were unidentified and the soil was root-adjacent, so a general root effect and host-specific recruitment remain open alternatives that host identification and root-attached sampling would separate. The genus contrasts in Figure 2d identify targets for culture, genome and plant-association studies.

Predicted pathways followed geography and compartment, suggesting that the taxonomic mosaic carries different metabolic potential. The dominant predicted biochemistry is aerobic heterotrophy: respiration through cytochrome *c* and the biosynthesis of amino acids, fatty acids and cofactors led the profiles, while carbon-fixation pathways and Cyanobacteriota were minor. This is the profile expected for an endospore-rich community that lives on scarce, episodic organic input rather than on photoautotrophic crusts [30]. The gene screen refines this picture: over a third of the recovered genomes carry the aerobic CO dehydrogenase and a fifth a group 1 [NiFe]-hydrogenase, mostly in Actinomycetota, and these carriers are most abundant in bulk shallow-subsurface soil and least abundant near roots. Oxidation of atmospheric H₂ and CO sustains desert soil bacteria between wetting events [37]; our data show that the genetic potential is widespread and concentrated where organic carbon input is smallest. The osmolyte genes also shift along the route, from trehalose synthesis in the west to ectoine and sporulation genes in the east where Bacillota rise, in line with the warmer and more alkaline conditions there. Along the route, most of the 92 pathways that changed monotonically were biosynthesis pathways that increased eastward (54 of 63), including unsaturated fatty-acid pathways, whereas near roots the supported differences included carbohydrate degradation and polyamine and thiamin biosynthesis. Agreement with matched shotgun profiles supports this broad reading. In the Namib, direct gas uptake, ATP and respiration measurements showed more than a 30-fold rise in hydrogen oxidation after wetting [37]. Our predictions identify where comparable process measurements are needed.

The relic-DNA experiment addresses DNA that remained accessible after cell membranes were damaged. PMA treatment reduced the expected richness in 8 out of 9 pairs (Figure 3c), showing that

PMA-sensitive DNA contributed taxa to these aliquots. PMA measures DNA accessible through damaged membranes [38]. A design balanced across the three compartments would test whether PMA-sensitive DNA is more abundant on the surface than below it.

The extraction blanks address a separate low-biomass risk: DNA introduced during laboratory processing. Candidate contamination was small in most linked samples, although one profile was strongly affected, and filtering left all 25 tracked conclusions unchanged (Figure 3d). This sensitivity supports the reported patterns for linked profiles [15]. The control design and its trip-specific limits are reported in Materials and Methods and Supplementary Section S2.

A regional framework for the Arabian Peninsula

Earlier Arabian studies established local habitat-specific patterns. A groundwater-fed UAE sabkha contained a saline microbial mat, while Saudi studies examined highland soils, arid locations, biological crusts, and desert plants [16, 17, 39, 13, 2]. This work revealed local habitat and plant effects without mapping the open interior. Our survey supplies that landscape frame and repeated compartment measurements.

The mean pH of 7.986, measured in a calcium chloride solution, is in the alkaline range reported for Saudi arid soils [17]. At AlUla, a plant-created patch of more fertile soil had a different bacterial community from nearby bare sand; the bare sand was also more alkaline [14]. In the Empty Quarter, pH shared substantial variation with geography. Plant patches may therefore modify a broader regional setting.

Regional experiments help interpret compartment and rainfall results. Around *Rhazya stricta*, the soil compartment had a larger effect than the simulated rainfall, while crust biomass increased 3-fold 1 week after watering [39]. At Jizan and Al Wahbah, the rhizosphere resembled the surrounding bulk soil while the root endosphere showed the strongest loss of richness and evenness and a composition that tracked host plant phylogeny [13]. Our near-root samples lie outside the compartments Eida et al. sampled; their lower evenness extends this local organization outward from the root. The rapid watering result independently shows that an Arabian desert community can change within days. Host identity, metabolite exchange, and plant benefit remain open questions.

The mapped variation makes this survey useful for regional decisions because a single reference site or compartment could be mistaken for the state of the wider desert. Saudi Arabia's cloud-seeding programme aims to enhance rainfall and support ecosystem restoration [40]. Our rainfall association suggests a concrete sampling schedule for testing soil-community change during the first several days after natural or enhanced rainfall. The western Empty Quarter also contains the Uruq Bani Ma'arid protected area and free-ranging Arabian oryx [41]; drought effects on reintroduced oryx show why habitat condition and climate matter for management [42]. Repeated microbial measurements could add soil and vegetation context before and after habitat restoration. For arid agriculture, the survey identifies comparable background sites and candidate taxa or predicted pathways for later plant trials [17, 13]. The linked samples, measurements, sequences and provenance make these comparisons reusable [21].

Limitations and identifying mechanisms

Our study measures associations. The sites lie on one west–east route, climate changes along that route, and every expedition traveled in the same direction. The shared route links location to collection order, time of day and changes during travel. Independent or crossing transects with randomized route direction would separate these effects from geography, but they are difficult in the roadless dune interior. The field team must use long, logistically feasible off-road traverses through an isolated landscape, which limits route randomization and simultaneous crossing surveys [1]. A next practical design could reverse the direction of travel between expeditions, add shorter crossing transects where terrain allows, and leave autonomous soil sensors at selected sites.

Several measurements needed to test ecological explanations are missing: soil microclimate (we used air measurements), soluble ions, conductivity and osmotic stress (XRF measures total elements [43]), and soil moisture and collection time (the rainfall model used gridded daily estimates and collection dates). Rainfall was too rare and uneven for the omission of one expedition to be a meaningful stability test; the randomization test instead shifted the timing of the rainfall observed within each expedition. Its first-several-days window remains an association, and models that allowed more complex geographic change weakened the evidence. Event-triggered sampling before and after storms, moisture sensors and watering experiments are needed to measure response time and test a mechanism. Root identity, root distance, carbon, texture and plant performance would likewise separate plant effects from shared habitat.

Shotgun metagenomics could recover genes without biases caused by 16S primers or by bacteria that carry different numbers of 16S gene copies. It could also link functions to genomes, resolve strains, and include archaea, fungi and viruses. It would test whether predicted pathways occur in the local gene pool, but DNA still measures potential. Transcripts, proteins, metabolites, cell counts and process rates are needed to show activity, while manipulations remain necessary for causation [44].

Control coverage was uneven. Whole-cell standards test extraction and purified-DNA standards enter after it, Trips 1–3 had no usable extraction blank (although co-prepared profiles were no more similar than distance-matched pairs; Supplementary Section S2), Trip 4 had no field-based positive or negative control (a PCR-stage-only positive control was run, and its PCR blank and no-template control were not sequenced), and Trip 5 controls used shotgun rather than 16S. Linked extraction blanks could train the contaminant sensitivity for Trips 4 and 5 only. Future work should sequence extraction and PCR blanks with every batch, use positive standards that enter at the same stage as the biological samples, and retain DNA concentrations.

Conclusions

The Empty Quarter is a structured microbial landscape. Bacterial assemblages turn over across the sand sea, nearby compartments differ in composition and evenness, environmental gradients track the broad pattern, and recent rainfall was associated with a short rise in richness. Predicted metabolic potential also changes with geography and compartment. These findings extend recurring dryland principles to the world's largest continuous sand desert while showing that the direction of diversity–environment associations remains local. The survey provides a regional reference and a set of testable hypotheses bounded by relic-DNA and contamination checks.

MATERIALS AND METHODS

Field design and sample description

We sampled a transect through the Empty Quarter during expeditions on March 17–20, 2023 (T1), July 14, 2023 (T2), February 17–21, 2024 (T3), August 19–21, 2024 (T4), and October 12–15, 2025 (T5). Each sampling campaign covered 60 sites, while extreme heat restricted T2 to 8 sites. At each site and visit, we sampled bulk surface soil (0 cm), bulk subsurface soil (5–10 cm) and subsurface soil referred to as “root-adjacent” collected near plant roots (5–10 cm), collecting three replicates per compartment. The sampling initially targeted the rhizosphere, but most plants were so desiccated that too little soil adhered to the roots; therefore, we collected the soil as close to the root as possible, which in some cases included the soil attached to the roots. We call this compartment “root-adjacent soil” because the soil was attached to roots in only some samples. Soil was placed in sterile tubes by scooping directly with the tube or with a sterile plastic spoon. Samples were kept at -20°C in a portable freezer during each expedition and transferred to -80°C in the laboratory.

Here, a profile means one quality-controlled 16S amplicon library and its counts for exact bacterial 16S amplicon sequences, which we call amplicon sequence variants (ASVs). Analyses that required repeat visits focused on sites 1–60 and used 1,227 profiles. Four locations sampled only during Trip 1 contributed 10 additional profiles. We retained them for description but excluded them from repeated-site analyses because later expeditions could not revisit these sites. Supplementary Table S1 reports the samples used in each analysis.

Amplicon processing

Between 0.9 and 1.2 g of soil was extracted with QIAGEN PowerLyzer PowerSoil or DNeasy PowerSoil Pro kits. Samples were homogenized with a PowerLyzer 24, processed manually or with a QIACube Connect, and eluted in 50 μL . The V3–V4 region of the 16S rRNA gene was amplified with the Bakt_341F (CCTACGGGNGGCWGCAG) and Bakt_785R (GACTACHVGGGTATCTAATCC) primers [45], both carrying Illumina overhang adapters. First-round PCR used TaKaRa Ex Taq DNA Polymerase (catalogue no. RR001B) in 50 μL reactions containing 5 μL of extracted DNA and 0.4 μM of each primer: 95°C for 3 min; 30 cycles of 95°C for 30 s, 63°C for 30 s and 72°C for 1 min; and a final 72°C extension for 6 min. Products were checked on a 1 % agarose gel; samples with an amplicon of the expected size (about 500 bp) were indexed, samples without a visible product were re-amplified with more DNA input, and samples that still failed were re-extracted.

Dual indices and Illumina adapters were added in a second-round PCR with the QIAGEN Multiplex PCR Kit (catalogue no. 206143) following the Illumina 16S protocol [46]: 94 °C for 15 min; 8 cycles of 94 °C for 30 s, 55 °C for 30 s and 72 °C for 30 s; and 72 °C for 5 min. Indexed libraries were purified with Agencourt AMPure XP beads, quantified with the Qubit high-sensitivity double-stranded DNA assay, sized on an Agilent TapeStation 4200 (D5000 ScreenTape), and pooled by concentration and size. Amplicon pools were sequenced on an Illumina NovaSeq 6000 on an SP flow cell with a 500-cycle reagent kit to generate 2 × 250-bp paired-end reads. Sample processing and library preparation were performed in the PI's laboratory and sequencing was conducted at the Bioscience Core Lab at KAUST.

Reads were processed with nf-core/Ampliseq v2.14.0 [47, 48]. We used DADA2 to infer exact amplicon sequence variants (ASVs) and SILVA 138.2 [49] to assign taxonomy. The complete count table retained all 351,472 ASVs and 24 control profiles from the Trip 5 sequencing run (18 extraction blanks and 6 no-template PCR controls). We excluded these controls and 10 biological profiles below 1,000 reads, leaving 1,237 profiles. Positive standards, PCR blanks and no-template controls were evaluated in their source-specific data and were never ecological profiles. The unfiltered biological table was the primary analysis input.

Assay controls and sensitivity analyses

We followed low-biomass guidance by analysing each control according to the stage it tested [15]. 17 sequenced extraction blanks could be linked by extraction day to 217 Trip 5 profiles. We pooled these blanks and marked an ASV as a candidate when it appeared in at least 2 blanks and was more prevalent in blanks than in biological samples in a one-sided Fisher test ($p < 0.1$). The 351 candidates were removed only from their linked profiles in a filtered copy used to test sensitivity. The remaining Trip 5 extraction blank and the 6 no-template controls had no extraction-day link and were characterized without being used to identify candidates. The 6 Trip 4 extraction blanks were sequenced on a separate run and denoised from the raw reads with the same pipeline; screened in the same way against the 95 Trip 4 profiles linked through the extraction-batch map, they gave 7 candidate ASVs (four unassigned at genus level, *Nesterenkonia*, *Brevibacterium* and *Alcaligenes*) whose removal changed a median of 0.00 % of reads per linked profile (maximum 14.3 %; 0.14 % pooled). Trips 1–3 had one extraction blank per extraction kit rather than per batch and no usable sequenced blank, so no extraction blank could be linked to their profiles (Supplementary Section S2). The Trip 4 PCR blank and no-template control were not sequenced because their index pairs conflicted with biological samples from Trips 1–3 on that run. No-template controls and PCR blanks contain the amplification reagents but no sample, so they test a later laboratory step; they were kept separate because their biological batch links were incomplete. Positive standards were used only to assess expected-taxon recovery and whether their expected taxa appeared in other samples. Supplementary Section S2 gives the trip-specific products, assay stages and remaining coverage gaps.

For the contamination sensitivity, we removed the 351 Trip 5 candidate ASVs only from the 217 linked Trip 5 profiles and the 7 Trip 4 candidates only from the 95 linked Trip 4 profiles. Trip 5 candidate sequences comprised 2.19 % of reads overall and a median of 0.40 % per profile (interquartile range, 0.16–0.99 %); 1 profile contained 56.60 % candidate reads (Figure 3d). Every filtered profile remained above 25,000 reads. We repeated 25 geographic, environmental and compartment conclusions with this filtered copy; all retained the same disposition as in the unfiltered primary analysis.

Relic-DNA removal and viability assessment

To distinguish DNA from intact, viable cells from total environmental DNA (including relic DNA), soil samples from 3 Trip 5 sites were subjected to PMAxx-based viability treatment, following established protocols for relic-DNA removal from soil matrices [50, 51]. For each site, 3 independent biological replicates were prepared as soil slurries by homogenizing soil in sterile phosphate-buffered saline (PBS, pH 7.4) at a 1:10 (w/v) ratio. Each replicate was partitioned into two aliquots: “treated” (T), used to quantify DNA from intact cells, and “untreated” (U), serving as a total-DNA baseline, yielding 9 matched treated/untreated pairs across the 3 sites. PMAxx dye (Biotium, Hayward, CA, USA) was added to treated aliquots at a final concentration of 50 μM, followed by a 10-minute dark incubation. Samples were photoactivated by exposure to a 450 nm LED lamp for 35 minutes, with tubes positioned on a white reflective surface 15–20 cm from the light source and rotated at 15-minute intervals. Immediately after photoactivation, all samples were stored at –20 °C until DNA extraction, which was performed on 860 μL of each sample using the DNeasy PowerSoil Pro Kit (QIAGEN). 16S rRNA amplicon libraries from these

samples were sequenced on an Illumina NovaSeq 6000 with the same 500-cycle kit and 2×250 -base paired-end configuration as the main sample set.

Propidium monoazide (PMA) enters cells with damaged membranes and prevents their exposed DNA from being amplified [38]. We compared 9 matched treated and untreated Trip 5 aliquot pairs. We calculated Hurlbert expected richness at 123,897 reads, the minimum depth among the 18 libraries, without random subsampling, and calculated Shannon diversity at full depth. Exact two-sided paired Wilcoxon tests compared the two endpoints; 99,999 bootstrap samples of the matched aliquots gave uncertainty for their mean differences (seed 20260805). Expected richness was lower after treatment in 8 pairs (mean treated-minus-untreated difference, -163.3 taxa; 95 % paired-aliquot bootstrap interval, -274.0 to -69.8 ; exact paired $p = 0.00781$; Figure 3c). This experiment measures a PMA-sensitive richness difference in these aliquots. Supplementary Sections S2 and S7.3 give the control provenance, quality checks and uncertainty calculations.

Environmental measurements

At each site visit we recorded air temperature and, where available, atmospheric pressure and relative humidity. We retained confirmed values and excluded one out-of-range measurement. We used these measurements in a secondary analysis (Supplementary Section S7.1) where we averaged each diversity measure within site and campaign, and tested its association with each variable. We included campaign and position along the transect (linear and quadratic terms) as covariates, with standard errors clustered by site to account for repeated visits. The nine tests were corrected together with the Benjamini–Hochberg method.

We obtained site-specific monthly air temperature, total rainfall, and relative humidity from the Open-Meteo historical API resource [26]. Each site had 49 complete months from January 2022 to January 2026. We averaged climate and the three diversity measures within each site, then used the 60 sites for Spearman correlation. We corrected the 9 climate–diversity tests together for multiple testing with the Benjamini–Hochberg method. For abundance, we selected the 200 genera with the largest mean abundance among those detected in at least 20 % of profiles. We added 0.5 to every count so that zero counts could enter the logarithm, then calculated each genus’s log ratio to the geometric mean abundance across all genera. This centred-log-ratio (CLR) transformation compares taxa within a relative composition instead of treating read counts as absolute abundances. We averaged the transformed abundances by site. We corrected the 600 climate–genus tests together in a second set.

We measured pH in 767 archived soil samples from all campaigns. Dry soil below 2 mm was mixed with 0.01 M CaCl_2 at 1:2.5 mass-to-volume and measured with an accumet AB315. We required a valid date and range, recorded calibration, an electrode slope of 95–102 %, and a pH 10 buffer read-back of 9.9–10.1. In total, 712 measurements passed and 702 matched ecology profiles, and we used measured values only. We averaged matched values within site, campaign and compartment. The composition test added pH after expedition, compartment and site and randomly rearranged the remaining variation within sites 999 times. We compared the transect result before and after pH adjustment in the same samples.

Laboratory XRF covered 725 archived-soil specimens. We retained 11 elements detected in at least 75 % of specimens, transformed values as $\log(1 + x)$, placed them on a common scale within each expedition and combined them by principal component analysis. This method constructs weighted combinations of the elements; the first component captures the largest shared change. We tested whether this component explained bacterial composition beyond expedition, compartment and site, using 999 random rearrangements within sites. Changes to the included taxa and the treatment of zero values tested sensitivity. XRF describes total elements [43]. Supplementary Sections S2–S4 give the laboratory quality checks, coverage and complete models.

Short-term rainfall response

We obtained daily precipitation for each site from the versioned NASA POWER PRECTOTCORR resource [25]; Open-Meteo provided an independent rainfall dataset for comparison. We excluded the sampling date because collection times were unavailable. Technical replicates were averaged within 617 site, expedition, and compartment groups.

We represented a temporary response by weighting rainfall over the 60 complete days before collection. For rainfall $r_{i,d}$ at site observation i and d complete days before collection, the pulse score for a possible

peak at day p was

$$x_i(p) = \sum_{d=1}^{60} r_{i,d}(d/p) \exp(1 - d/p).$$

The weight rises to one at $d = p$ and then declines. We searched $p = 0.5, 1.0, \dots, 30.0$ days for expected richness at 25,000 reads, Shannon diversity and normalized evenness. Each model allowed each site and compartment to have its own baseline, each expedition to have its own mean, and longitude and collection day to have separate steady trends within each expedition.

To account for choosing the strongest result from many peaks and diversity measures, we shifted the full 60-day rainfall sequence in a circle 19,999 times, independently within each expedition (seed 20260804). Each shifted dataset retains the amount, rarity and spatial pattern of rainfall but changes its timing relative to collection. We used a one-sided test for the stated prediction of a positive pulse and also report a two-sided test that allows either direction; both tests cover the complete search. Confidence intervals for the selected effect used sandwich covariance clustered by site. We resampled whole sites 9,999 times (seed 20260805) to estimate the share of remaining variation explained by rain and the uncertainty in peak selection. Supplementary Section S7 describes these calculations and the comparison of rainfall datasets. Additional analyses changed the rainfall dataset, contaminant filtering, pH adjustment, and geographic trend and included samples. To check the assumed pulse shape, we fitted rainfall from 1, 2, 3–4, 5–7, 8–14, 15–30 and 31–60 complete days before collection together for each rainfall dataset, with site-clustered uncertainty. We also used future rainfall as a negative control. Supplementary Section S7 gives these results, and the exact reproducibility manifests.

Community, diversity and spatial analyses

To compare richness at the same sequencing depth, we calculated Hurlbert's expected richness: the number of taxa expected if each profile contributed 25,000 reads. We fixed this depth before analysis as a common standardization depth near the 5th percentile of library size (median 382,432 reads per profile), so that expected richness is comparable across profiles while 95 % of them are retained; the 60 profiles with fewer reads (55 of them core profiles) contributed no expected-richness value, and no profile was discarded on this basis (Supplementary Section S2). Shannon diversity used full depth. We averaged profiles within site, expedition and compartment before paired compartment comparisons. Means and intervals came from repeatedly sampling whole sites with replacement. We drew 1,000,000 whole-site bootstrap samples (seed 20260723), keeping all repeated observations from one site together. We corrected each family of 3 contrasts. An additional evenness analysis used $H / \log(E[S_{25k}])$, where H is Shannon diversity and $E[S_{25k}]$ is expected richness at 25,000 reads. This measures abundance balance relative to common-depth richness. Supplementary Section S5.2 gives campaign-specific contrasts and correction families.

For composition, we combined replicate counts within expedition, site and compartment. We selected the 200 most abundant genera among those detected in at least 20 % of profiles, added 0.5 to every count to handle zeros, and used the CLR transformation defined above [23]. This small addition is called a pseudocount. The paired test used 170 complete three-compartment comparisons across 60 sites. Pairwise tests randomly reversed the direction of each within-pair difference while keeping all observations from a site together. Alternative taxon sets, treatments of zero counts and expedition omissions tested sensitivity.

For broad spatial structure, we retained core-site groups with at least 2,000 reads, removed the CLR mean for each expedition and compartment, and averaged within site. The route model allowed both steady change from west to east and a smooth change in rate that could reverse once, represented by straight-line and squared-location terms. It used the same 200 genera and 999 random rearrangements of whole sites. Taxon sets, trend shapes and expedition omissions tested sensitivity. To quantify sampling uncertainty in the explained share, we removed each of the 60 sites in turn, refitted the model, calculated the delete-one-site jackknife standard error, and formed a 95 % t interval with 59 degrees of freedom.

To measure distance decay within each compartment, we removed the campaign mean from the CLR composition at each of the 60 sites and related Aitchison dissimilarity to distance. We randomly reassigned location labels to whole sites 9,999 times across all compartments, so the site was the unit of resampling. For comparisons based only on whether a taxon was present, we pooled counts within site and compartment and randomly subsampled each pooled sample to the same 12,865 reads (seed 20260728). We partitioned Sørensen dissimilarity into taxon replacement and nestedness [24]. Delete-one-site jackknife intervals for the distance slopes and mean replacement shares removed every pair involving the omitted site.

Supplementary Section S5 gives the spatial prediction, ASV-resolution and dispersion diagnostics.

To describe the taxa behind these patterns, we summed ASV counts per phylum, class and genus and expressed them as shares of total reads per profile across the 1,227 core profiles, overall and by compartment and by transect third (20 sites each, ordered by route position). For the 200 genera of the route model, we correlated the site-level CLR values (expedition-by-compartment means removed) with route position (Spearman; Benjamini–Hochberg over 200 tests) and estimated the east-minus-west difference in mean CLR with a delete-one-site jackknife interval. To compare within-transect with between-desert differences under one rule, we subsampled every Empty Quarter profile and every Atacama pit profile [12] 20 times to 8,000 reads (seed 20260903), called a genus detected in a set when it occurred in at least 10% of subsampled profiles, and computed Jaccard overlaps between transect thirds and between the Empty Quarter and the pit (all depths, and 2.5–10 cm only). This comparison uses presence only, across datasets that differ in primer (V3–V4 versus V4) and DNA fraction (total versus intracellular). Site-level pH and climate means were correlated with route position to state the direction of the gradients. These summaries are descriptive and add no claim to the tracked conclusions (`analysis/v3/taxon_context/`).

For each of the 200 genera we also tested the 3 paired compartment contrasts in the same block-centred site tensor as the composition test (sign-flip over sites; Benjamini–Hochberg over 600 tests; site bootstrap intervals), correlated site-level CLR values with site-mean pH and with the laboratory-XRF elemental axis (Spearman; Benjamini–Hochberg over 200 tests each), and compared the 44 sand-dune and 9 saline-pan sites, using the field landform recorded for each site, with a label-permutation pseudo- F and per-genus Mann-Whitney tests, without and with adjustment for route position. The route model, the genus route correlations and the climate-diversity correlations were refitted on the dune sites. Genus occupancy per compartment used counts pooled across expeditions within site and subsampled 20 times to 12,865 reads (seed 20260904; `analysis/v3/biology_context/`).

Predicted functional potential and shotgun comparison

PICRUSt2 v2.4.1 predicted KEGG Ortholog groups and MetaCyc pathways from the ASV tables [27]. We matched 1,227 core-site profiles, summed replicates within expedition, site and compartment, and ranked pathways detected in at least 20% of groups. The primary analysis used the 200 with largest mean abundance, a 0.5 pseudocount and CLR coordinates. For geography, we removed expedition-by-compartment means and tested the route model that allowed the rate of change to vary smoothly and reverse once, using 9,999 random rearrangements of whole sites. The pathway geography interval used the same delete-one-site jackknife and 95% t interval as the taxonomic route model. For compartment, we expressed each complete three-compartment comparison relative to its own mean and then averaged within site. The overall test randomly reassigned compartment labels within sites; pairwise tests randomly reversed the site-level differences. We corrected 3 profile comparisons together and 600 pathway comparisons together. Pathway sets, pseudocounts and expedition omissions tested sensitivity. To describe the dominant predicted metabolism, we expressed each pathway as a share of predicted pathway abundance per profile, averaged across the 1,227 core profiles, and grouped pathways by keywords in their MetaCyc descriptions into biosynthesis, degradation or utilization, energy and central metabolism, and other; the rules and the ranks of the carbon-fixation pathways are archived with the results. For a fixed list of 20 trait genes (KO identifiers in the repository), we counted carriers among the 975 annotated genomes of the companion catalogue (eggNOG-mapper KO annotations), weighted them by CoverM genome abundance in the 150 shotgun libraries, and summed the predicted abundance of the same KOs in the PICRUSt2 profiles as a share of all KO abundance. Compartment contrasts used site-level paired $\log_{10}$ ratios with sign-flip tests (Benjamini–Hochberg over 60 tests), route correlations used Spearman over site means (20 tests), and agreement between the two sources used Spearman over 119 matched libraries (`analysis/v3/trait_genes/`).

For each predicted profile, the weighted nearest-sequenced-taxon index (NSTI) summarized the distance between its ASVs and sequenced reference genomes. A smaller value means that the prediction requires less extrapolation. For an empirical comparison, we matched PICRUSt2 and shotgun-derived abundance for 3,471 KEGG Ortholog gene families in 125 samples, calculated a Spearman rank correlation within each sample, and also correlated the two community-average profiles. The shotgun data come from a separate, ongoing metagenomic study of the same expeditions (BioProject [\[TODO: BioProject accession\]](#)), in which 125 shotgun libraries from Trip 5 matched 16S profiles in this survey. That study's

assembly, binning, dereplication and eggNOG-mapper v2.1.12 annotation of the genome catalogue produced the CoverM genome-abundance profiles and KEGG Ortholog annotations we reused here; its results will be reported separately. Supplementary Section S8 reports the uncertainty method and the limits of pathway and marker-gene predictions.

Cross-desert contextual analysis

We compared effects in 2 public Atacama datasets without merging their ASV counts. In QIITA study 10360 (PRJEB17617), technical replicates were averaged within 16 sites; Spearman correlations related diversity to soil humidity before and after adjustment for latitude, longitude and elevation. In PRJEB39249, 62 intracellular-DNA profiles were aggregated to 23 sampled depths. Three depth zones were compared in 200-genus CLR coordinates with 9,999 label rearrangements; diversity was averaged after subsampling each profile 100 times to 8,000 reads. Supplementary Section S9 gives the upstream processing and comparison limits.

Sensitivity analyses and reproducibility

To test whether sequencing depth or recorded extraction method explained the Shannon compartment differences, we fitted a Gaussian generalized estimating equation with sequencing depth, expedition and extraction protocol. This regression method accounts for repeated observations from the same site. We treated these models as robustness checks because extraction protocol and laboratory batch were incompletely recorded and partly linked to expedition. The companion data descriptor covers the field, specimen, amplicon, environmental, XRF and knowledge-graph resources [21]. Inputs, scripts, machine-readable results, random seeds and checksums are maintained in the dedicated reproducibility repository at <https://github.com/bio-ontology-research-group/empty-quarter-ecology-reproducibility>. The repository links each analysis to its exact input, machine-readable result, random seed and checksum, and links shared data products to the companion resource. Public deposition of the ecology package and upstream amplicon, PMA, shotgun-read and genome resources remains a submission requirement.

Software

Amplicon reads were processed with nf-core/Ampliseq v2.14.0 (Nextflow; DADA2 and SILVA 138.2) on the KAUST Ibox cluster, where the eggNOG-mapper v2.1.12 annotation of the shotgun genome catalogue also ran. All statistical analyses were run on a Linux workstation in Python 3.11.14 with pandas 2.2.3, NumPy 2.1.3, SciPy 1.14.1, statsmodels 0.14.6, scikit-learn 1.5.2, scikit-bio 0.7.3 and Biopython 1.84; PICRUSt2 v2.4.1 predicted functional potential. Figures were drawn with Matplotlib 3.9.4 (FreeType 2.14.3). The complete environment is pinned as a conda lock file with hashed pip requirements in the reproducibility repository, and every reported number is regenerated from the committed inputs by the repository's tests. We used Claude (Anthropic) to assist with writing analysis code and with editing the manuscript; all analyses, results and claims were checked and verified by the authors.

FUNDING

SMA was supported by the Fonds National de la Recherche (FNR), Luxembourg (Project Ref. 18047750).

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